DRUGS · THYROID AND PARATHYROID

Carbimazole and propylthiouracil

Carbimazole and propylthiouracil (PTU) block thyroid hormone synthesis. They are first-line treatment for Graves' disease and are used to make thyrotoxic patients euthyroid before thyroidectomy, because operating on an uncontrolled thyrotoxic patient risks thyroid storm. Their most important side effect is agranulocytosis, which presents with a sore throat and fever: an ENT emergency that is easily missed.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

CARBIMAZOLE AND PROPYLTHIOURACIL

	SUMMARY
Agents	Carbimazole (prodrug of methimazole); propylthiouracil
How they work	Inhibit thyroid peroxidase, blocking iodination of thyroglobulin and coupling of iodotyrosines; PTU also blocks peripheral conversion of T4 to T3
ENT uses	Making Graves' disease and toxic nodular goitre euthyroid before thyroidectomy; definitive medical treatment of Graves' disease (12 to 18 months)
Carbimazole dose	20 to 40 mg daily, titrated to free T4 (titration regimen), or 40 mg daily with levothyroxine (block-and-replace)
PTU dose	200 to 400 mg daily in divided doses, then reduced; preferred in the first trimester of pregnancy and in thyroid storm
Monitoring	Thyroid function every 4 to 6 weeks; full blood count if any symptoms of infection; liver function with PTU

	SUMMARY
Pregnancy	PTU in the first trimester (carbimazole embryopathy), carbimazole thereafter

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Preparing for thyroidectomy

Aim for a euthyroid state on carbimazole (with propranolol for symptoms) before elective surgery. When antithyroid drugs cannot be used or the patient needs urgent surgery, prepare with potassium iodide, beta blockers, steroids and sometimes cholestyramine under endocrinology (see preoperative preparation for thyroid surgery).

3 CHAPTER THREE Safety

What goes wrong.

3.1 Agranulocytosis

Occurs in about 0.1% to 0.5%, usually in the first 3 months. Every patient must be told to stop the drug and have an **urgent full blood count** if they develop a sore throat, mouth ulcers, fever or other signs of infection. In a patient with a sore throat on carbimazole, check the neutrophil count before diagnosing tonsillitis, and treat neutropenic sepsis urgently.

3.2 Other adverse effects

Rash and pruritus (common), arthralgia, raised liver enzymes; PTU causes rare but severe **hepatotoxicity** (avoid it as long-term treatment); ANCA-associated

vasculitis with PTU; carbimazole in early pregnancy causes aplasia cutis and other embryopathy.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism and the difference between carbimazole and PTU.
- **Agranulocytosis** and the sore throat warning.
- Making the patient euthyroid before surgery.

4.2 Common pitfalls

COMMON PITFALLS

- Treating a sore throat in a patient on carbimazole as tonsillitis without a blood count.

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